

TASK 3 REPORT

Abstract — This report presents a 24-configuration ablation study intended to close the performance gap identified in Task 2, followed by 5-fold subject-wise cross-validation and a significance test of the resulting final GNN against a Random Forest baseline. **A critical scoping note precedes the results:** this notebook re-derives its dataset under a materially different protocol than Tasks 1–2 (EMG-only, 10 channels vs. EMG+glove 32; 20-sample/200 ms windows at 0% overlap vs. 200-sample/2 s at 50%; Exercise 1 only vs. all three pooled; 13 classes including rest vs. 23 excluding rest). Ablation results and the final CV comparison are therefore internally consistent but **not directly comparable to the Task 2 baseline table**. Within its own protocol, the best ablation configuration (SAGE architecture, 1 GCN layer, no dropout, mean pooling, anatomical adjacency graph) still loses to a Random Forest baseline in Macro-F1 across all 5 cross-validation folds (0.1470 ± 0.0074 vs. 0.2238 ± 0.0369); the difference is directionally consistent but not statistically significant at $\alpha = 0.05$ (Wilcoxon $p = 0.0625$, $n = 5$ folds). The explainability component (SHAP + LIME) required for this task's deliverable was not present in the supplied materials and is flagged as outstanding.

Index Terms — ablation study, GNN, cross-validation, significance testing, explainability

I. Introduction

Task 2 established that an initial correlation-graph GCN substantially underperforms classical baselines (Macro-F1 0.3925 vs. SVM's 0.7842). This report was intended to systematically vary architectural and training choices to close that gap. However, the ablation notebook supplied for this task uses a reconstructed dataset with a different channel set, window length, exercise subset,

and class count than Tasks 1–2 (see Section II). This is documented transparently below rather than presented as a seamless continuation.

II. Deviation from the Task 1/2 Protocol — Documented Explicitly

Parameter	Task 1 / Task 2	Task 3 (as supplied)
Channels (nodes)	32 (10 EMG + 22 glove)	10 (EMG only)

Parameter	Task 1 / Task 2	Task 3 (as supplied)
Window length	200 samples (2.0 s)	20 samples (0.2 s)
Overlap	50%	0%
	(100-sample step)	(20-sample step)
Exercises used	All 3, pooled	Exercise 1 (A) only
Rest class	Excluded	Included
Classes	23	13 (12 gestures + rest)
Node features	RMS, MAV, WL, VAR (4)	MAV, RMS, WL, ZC, SSC (5)
Windows	30,936	133,486
Train/test split style	Fixed subject blocks (1–20 / 21–27)	GroupShuffleSplit (random subject grouping, seeded)

Consequence: the "baseline to beat" used later in this report (a freshly trained Random Forest under Task 3's own protocol, Macro-F1 0.2238) is not the same baseline established in Task 2 (SVM, Macro-F1 0.7842) — the two numbers describe different classification problems (13-class single-exercise-with-rest vs. 23-class three-exercise-no-rest) and must not be compared directly. This report treats them as separate reference points.

III. Ablation Setup

Fixed subject-wise held-out split (train/val/test), fast-mode training (15 epochs, patience 5) across a single configurable GNN (ConfigurableGNN, supporting GCN/GAT/GraphSAGE) with a default configuration of: architecture GAT, hidden dim 64, 2 layers, dropout 0.3, mean+max

pooling, weighted correlation-based edges, AdamW, LR scheduler on, class-weighted loss.

Twenty-four experiments varied one axis at a time from this default: number of layers (1/2/3), hidden dimension (32/64/128), dropout (0.0/0.2/0.5), graph type (correlation vs. fixed anatomical adjacency — 8-electrode ring + 2 muscle-specific electrodes linked to the two nearest ring nodes), architecture (GCN/GAT/SAGE), edge weighting (on/off), pooling

(mean/max/mean+max), and optimizer/scheduler/loss-weighting toggles.

Note on requirement coverage: the assignment's ablation list also specifies batch normalization and early stopping as axes to vary. BatchNorm is present in every configuration (not toggled off in any run) and early-stopping patience is fixed per phase (5 in ablation, 12 in final training) rather than swept as its own axis. These two axes were therefore not independently ablated in the supplied notebook.

IV. Ablation Results (selected, full 24-row table in ablation_results.csv)

Experiment	Config	Accuracy	Macro-F1
A1_layers_1	num_layers=1	0.4893	0.1396 (best)
A5_arch_SAGE	arch=SAGE	0.4887	0.1339
A7_pool_mean	pooling=mean	0.4997	0.1341
A1_layers_3	num_layers=3	0.5018	0.1333
Baseline GNN	(defaults)	0.4831	0.1269
A4_graph_adjacency	graph=adjacency	0.4908	0.1276
A4_graph_correlation	graph=correlation	0.4822	0.1255
A5_arch_GAT	arch=GAT	0.4865	0.1191
(default)			(worst arch)
	weighted_loss=false		0.1090 (worst overall Macro-F1)
A8_unweighted_loss		0.5499	

Per-axis findings: (1) fewer GNN layers perform better ($1 > 3 > 2$), suggesting over-smoothing on this small 10-node graph; (2) hidden size has little effect (64 best); (3) less dropout is better ($0.0 > 0.2 > 0.5$); (4) anatomical adjacency marginally beats correlation-based graphs, though both are close to baseline; (5) GraphSAGE outperforms both GCN and GAT (the default architecture, GAT, is the *worst* of the three); (6) weighted edges beat unweighted; (7) simple mean pooling beats mean+max and max; (8) class-weighted loss is important — removing it produces the highest raw accuracy (0.55) but the worst Macro-F1 (0.109), a textbook illustration of why

accuracy alone is misleading under class imbalance (as flagged in the Task 1 EDA).

V. Final Configuration Selection

Per-axis best configurations were merged into the final model only where they beat the "Baseline GNN" row's Macro-F1 (0.1269): architecture $\rightarrow$ SAGE, layers $\rightarrow$ 1, dropout $\rightarrow$ 0.0, pooling $\rightarrow$ mean, graph $\rightarrow$ adjacency; edge-weighting, optimizer, scheduler, and loss-weighting were retained at default values because no tested alternative in those axes beat the baseline. Final configuration: {arch: SAGE, hidden_dim: 64, num_layers: 1, dropout: 0.0, pooling: mean, edge_weighted: true, graph_type: adjacency, optimizer: AdamW, use_scheduler: true, weighted_loss: true, epochs: 80, patience: 12}.

VI. 5-Fold Subject-Wise Cross-Validation

GroupKFold (5 folds) on 27 subjects, with per-fold train/val/test subject disjointness explicitly asserted (no leakage).

Metric	Final GNN (mean $\pm$ std)	Random Forest baseline (same folds, mean $\pm$ std)
Accuracy	0.4186 $\pm$ 0.0627	0.5978 $\pm$ 0.0364
Macro-F1	0.1470 $\pm$ 0.0074	0.2238 $\pm$ 0.0369
Weighted-F1	0.4582 $\pm$ 0.0435	—
Precision (macro)	0.1633 $\pm$ 0.0079	—
Recall (macro)	0.1634 $\pm$ 0.0123	—

Per-fold Macro-F1 (GNN vs. RF): Fold 1: 0.1562 vs. 0.2229 (0.0667); Fold 2: 0.1355 vs. 0.1622 (0.0267); Fold 3: 0.1470 vs. 0.2297 (0.0827); Fold 4: 0.1483 vs. 0.2504 (0.1021); Fold 5: 0.1478 vs. 0.2540 (0.1062). The GNN is worse than the Random Forest baseline in every single fold.

VII. Significance Test

Wilcoxon signed-rank test, paired per fold, on Macro-F1: statistic = 0.0000, $p = 0.0625$ ($= 0.05$) $\rightarrow$ **not statistically significant**,

though this reflects the minimum possible test statistic for $n = 5$ paired folds where one condition is uniformly worse in every fold; with only 5 folds the test lacks power to confirm significance despite a perfectly consistent direction of effect. This should be reported as "consistent but not confirmed significant at $n=5$," not as "no difference."

VIII. Final Comparison Table

Model	Accuracy	Macro-F1
Best Baseline (Random Forest, Task-3 protocol, fold estimate)	0.5978	0.2238
Final	0.4186	0.1470
Improved GNN (5-fold CV mean)		

The Task 2 initial-GNN metrics file (`gnn_metrics.csv`) referenced by this notebook was not found at its expected path in this run, so the "first GNN" row could not be merged into this table programmatically; it is included qualitatively via cross-reference to the Task 2 report (Macro-F1 0.3925 under the *different*, EMG+glove/23-class protocol — not directly comparable to the figures in this table).

IX. Explainability (SHAP + LIME)

Not available in the supplied materials. The assignment requires SHAP and LIME explanations for one correctly and one incorrectly classified sample, delivered via a separate `..._task3_explainability.ipynb` notebook. No such notebook was provided for this report. This

section is a placeholder and must not be filled with fabricated explanations; populate it once that notebook exists.

X. Discussion

Two findings dominate this task. First, the ablation sweep, despite covering seven of the nine requested axes (batch norm and early-stopping were not independently varied — Section III), never produces a configuration competitive with even a same-protocol Random Forest baseline — the gap between GNN and RF actually widens slightly under 5-fold CV (Macro-F1 ratio $0.66\times$) compared to the single-split ablation estimates. Second, the protocol drift documented in Section II means this task's numbers cannot be read as "the same model, improved" relative to Task 2; they describe a different, narrower classification problem (single exercise, EMG-only, rest included). Both facts should be stated explicitly in any presentation of this work, since a reviewer comparing Task 2's 0.78 SVM Macro-F1

to Task 3's 0.15–0.22 range without this context would reasonably conclude something is broken, when the actual explanation is a scope change plus a GNN that has not yet matched classical baselines on either version of the task.

XI. Conclusion

Under its own (narrower) protocol, the best-ablated GNN configuration (GraphSAGE, 1 layer, no dropout, mean pooling, anatomical-adjacency graph) does not outperform a Random Forest baseline; the difference is consistent in direction across all 5 CV folds but not statistically significant at $n = 5$. The GNN approach, across both Task 2 and Task 3 protocols, has not yet demonstrated an advantage over classical feature-based baselines on NinaPro DB1 gesture classification. Outstanding work: (1) reconcile the Task 3 protocol with Task 1/2's scope before final submission, or explicitly justify the narrower scope as an intentional refinement; (2) complete the SHAP/LIME explainability notebook; (3) consider whether the anatomical-adjacency and correlation graphs tested so far are the right candidate space, given that neither meaningfully beat the other or the model's own baseline.

Deliverables: code/task3/Group04_NinaProDB1_task3_improvement_ablation.ipynb; code/task3/..._task3_explainability.ipynb (missing); models/; report/task3/Group04_NinaProDB1_task3_report.pdf (final).